

Recipe

Android Mobile Application – Project Proposal

Project Name: Recipe

Platform: Android (Kotlin + Jetpack Compose)

Backend / Data Source: TheMealDB API

Document Type: Project Proposal

Date: August 2026

1. Introduction

Recipe is a mobile Android application designed to help users discover and organize meals using real recipe data from the internet. Similar to how movie apps use TMDb as a database, this project uses **TheMealDB** as a free online recipe API to provide meal names, images, categories, and ingredients.

The app focuses on a practical everyday use case: finding recipes quickly, saving favorites, planning a shopping list, and later using AI chat to support cooking decisions.

2. Problem Statement

Many people struggle to decide what to cook or forget which ingredients they need. Existing recipe websites are often cluttered and not optimized for fast mobile use. Users need a simple app that can:

- Browse and search recipes easily
- Show clear food images and ingredient information
- Save favorite recipes
- Generate or manage a shopping list
- Offer smart help through chat / AI in future versions

3. Project Objectives

- Build a clean and modern Android recipe application
- Integrate a public recipe API (TheMealDB) as the backend data source

- Implement a home screen with search and recipe cards
- Add navigation for Home, Chat, Shopping List, Favorite, and Profile
- Provide a branded splash screen with Lottie animation
- Prepare the foundation for AI features and personal kitchen inventory

4. Proposed Solution

The app will present recipes in a warm, food-focused design with a cream background and a salad-bowl logo. Users will open the app through an animated splash screen, then land on the home page where they can search and browse meals fetched from TheMealDB.

A bottom navigation bar will give access to the main sections of the app, with **Shopping List** placed in the middle for quick access.

5. Main Features

Feature	Description
Splash Screen	Animated Lottie food intro with app name and branding
Home Page	Recipe grid, logo, search bar, and favorite hearts
TheMealDB Integration	Fetch and search real recipes with images and ingredient counts
Bottom Navigation	Home, Chat, Shopping List, Favorite, Profile
Favorites	Mark recipes as favorite
Shopping List	Create a list of missing ingredients for selected recipes
Chat / AI Assistant	Suggest recipes from ingredients and answer cooking questions
Profile	User settings and personal preferences

6. Design Overview

- **App name:** Recipe
- **Logo:** Static salad-bowl image (also used as launcher icon)
- **Colors:** Cream background, terracotta accents, forest green text, sage green details

- **Home layout:** Title on the left, logo on the top right, search under the header, 2-column recipe cards
- **Bottom bar:** Home · Chat · List · Favorite · Profile

7. Technical Stack

Area	Technology
Language	Kotlin
UI	Jetpack Compose + Material 3
Architecture	UI / ViewModel / Repository / API layers
Networking	Retrofit + OkHttp + Gson
Images	Coil
Animation	Lottie
API / Backend	TheMealDB (https://www.themealdb.com/api.php)
IDE	Android Studio

8. Why TheMealDB?

TheMealDB works as a public recipe database in a similar way to TMDB for movies. It provides endpoints to search meals, load meal details, and get images without requiring a complex custom backend at the beginning of the project. This allows faster development and a realistic online data experience.

9. Future Enhancements

- Recipe details screen with full instructions and ingredient list
- Persistent favorites using local database (Room)
- Shopping list generation from selected recipes
- AI chat for recipe suggestions and substitutions
- User authentication and personal profile

10. Expected Outcomes

By the end of the project, the result will be a working Android application that demonstrates API integration, modern UI design, navigation, and practical mobile development skills. The app will be useful for everyday cooking.

11. Conclusion

Recipe combines a clean visual design with real online recipe data to create a simple and useful cooking companion. Using TheMealDB as the backend makes the project practical, scalable, and similar in concept to popular content apps that rely on public APIs.